

ENME608: Mechanism and Dynamics of Machinery

Lecture 2: Introduction to Linkages

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Assessment

Assessment Type	Weighting	Due Date
Mid-semester Test	25%	In Week 6 (Date & Time: TBA)
Assignment	25%	31 May 2026, Online Submission by 11:59 PM
Final Controlled Assessment	50%	After Week 12 (Date & Time: TBA)

Contents of ENME608

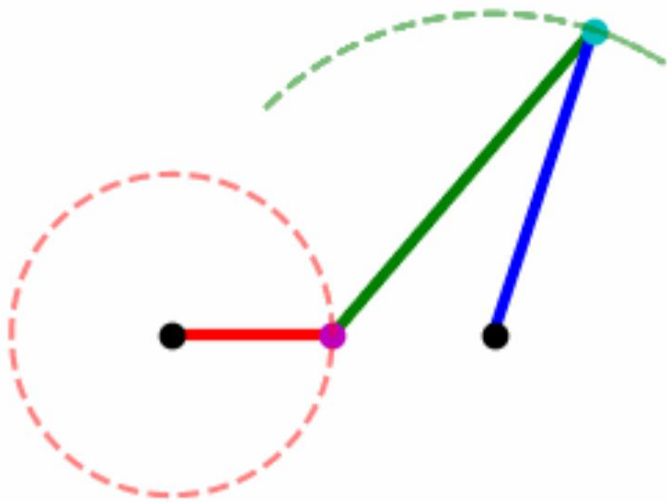
Mid-Sem Break	Mechanism and Machine Concepts	Week 1	Contents of Mid-Semester Test
	Linkages	Week 2	
	Position Analysis	Week 3 – 4	
	Velocity Analysis	Week 5	
	Acceleration Analysis	Week 6	
	Cam	Week 7	Contents of Final Exam
	Cam (continued) and Gear	Week 8	
	Gear (continued)	Week 9	
	Dynamics Fundamentals	Week 10	
	Force Analysis	Week 11	
	Force Analysis (continued) and revision	Week 12	
Final Exam (TBA)			

Reference

- Recommended Textbook
 - Robert L. Norton, *Design of Machinery : An Introduction to the Synthesis and Analysis of Mechanisms and Machines*, 6th Edition, McGraw Hill, 2019, ISBN-10: 1260113310 , ISBN-13: 978-1260113310
(Online version available: AUT Library or
ENME608: Course Resources in Canvas)

Four bar linkage

- One of the simplest examples of a single DOF constrained linkage. It consists of **crank**, **rocker** and **coupler** (connecting rod).



Crank-Rocker Mechanism

_____ is a link which makes a complete revolution and is pivoted to ground.

_____ is a link which has oscillatory (back and forth) rotation and is pivoted to ground.

_____ is a link which has complex motion and is not pivoted to ground.

Grashof condition

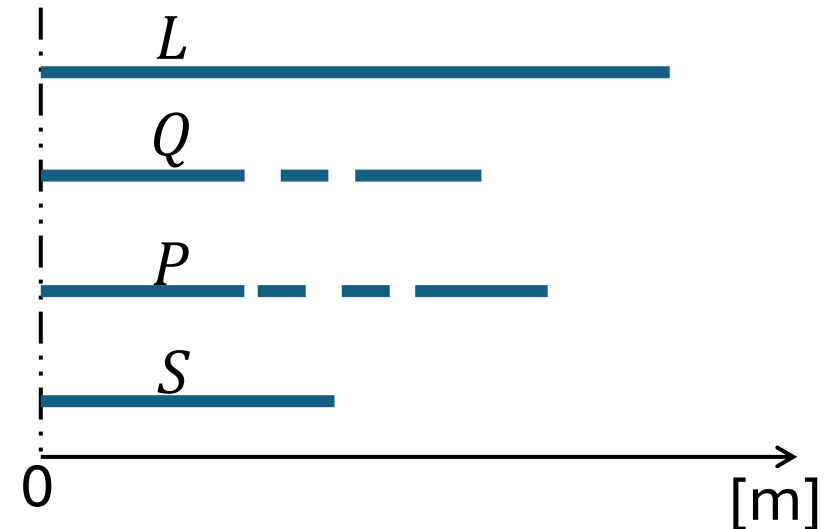
- Generation of various forms of motions is governed by the **Grashof condition** which predicts the rotation behaviour or **rotatability** of the linkage based only on the link lengths (S , L , P and Q).

S : length of shortest link

L : length of longest link

P : length of one remaining link

Q : length of other remaining link



$L < (S + P + Q) \Rightarrow$ Four links can be assembled.

$L = (S + P + Q) \Rightarrow$ Four links can be assembled but cannot move.

Grashof condition (Cont.)

At least one link can make a full revolution w.r.t. the ground plane.

Grashof

Class I kinematic chain

$$S + L < P + Q$$

Class III* kinematic chain

$$S + L = P + Q$$

No link can make a full revolution w.r.t. any other link.

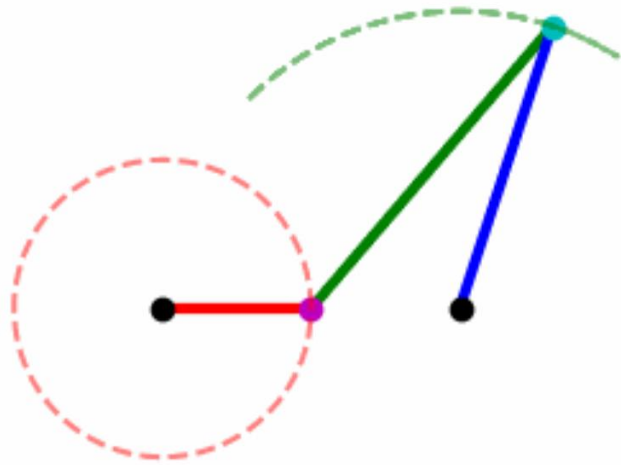
Non-Grashof

Class II kinematic chain

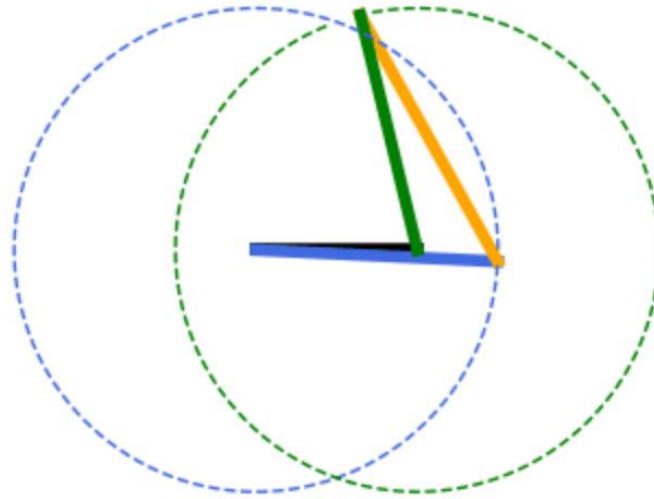
$$S + L > P + Q$$

* Special-case Grashof

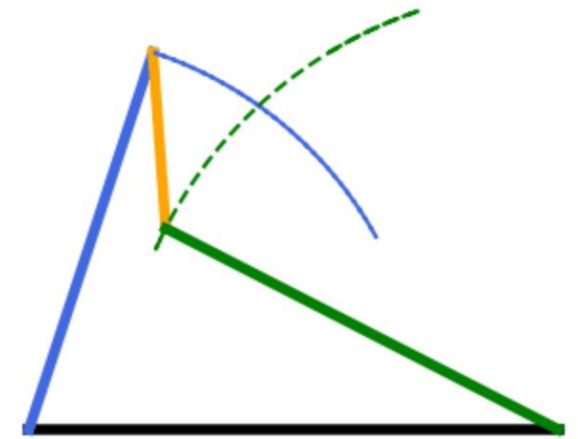
Grashof condition – **Class I (Grashof)**



Crank-Rocker

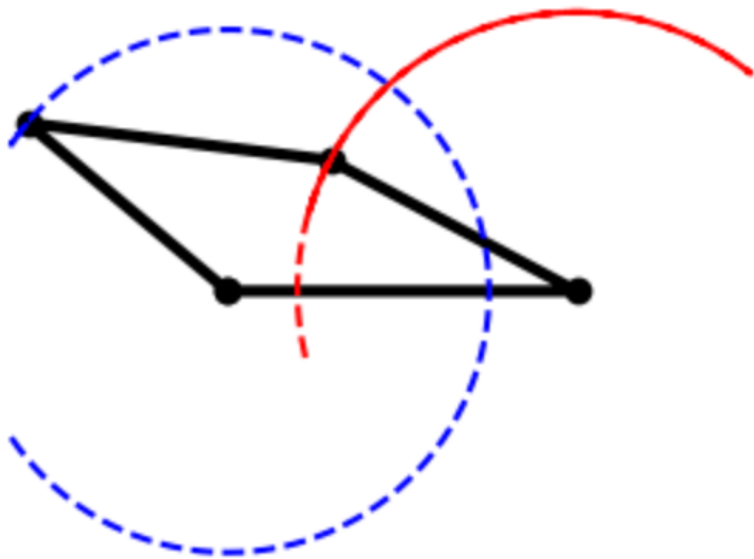


Double-Crank



Grashof Double-Rocker

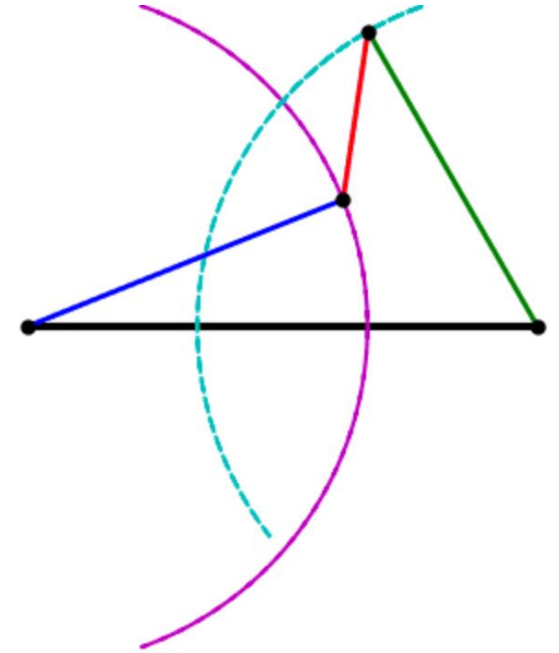
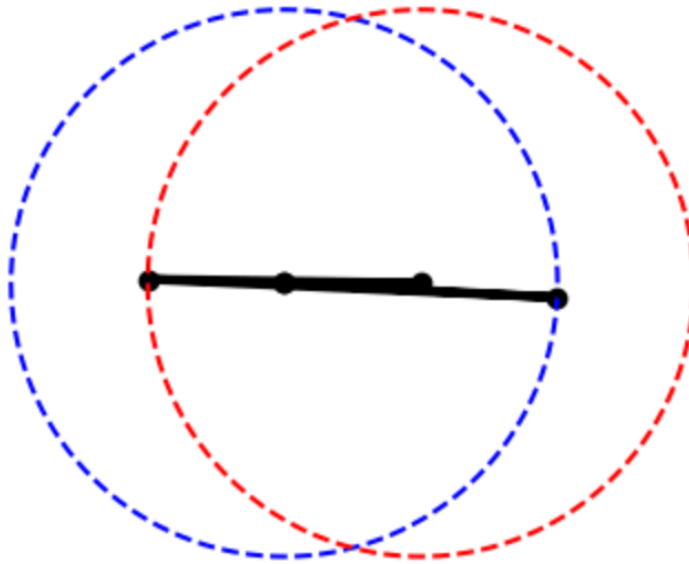
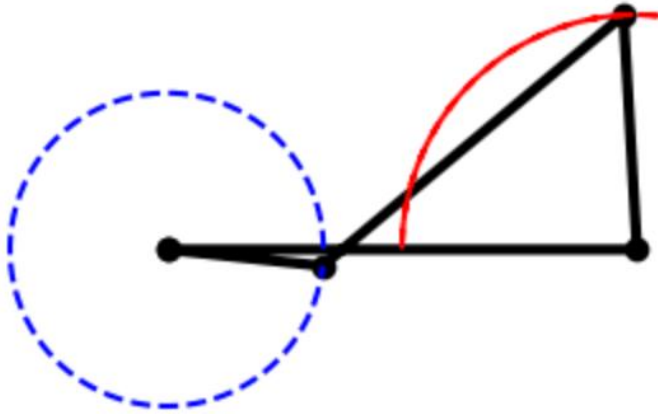
Grashof condition – **Class II (Non-Grashof)**



Triple-rocker

All inversions of Class II kinematic chains are Triple-Rocker.

Grashof condition – **Class III** (Special-case Grashof)



The links can become **collinear** and the driven (output) link can change direction of rotation – **uncertainty configuration**.

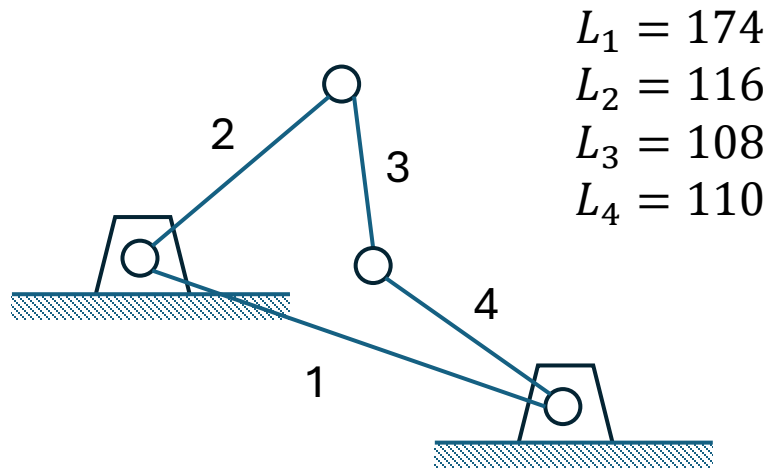
Barker's Complete Classification of Planar Fourbar Mechanisms

Referred from Page 61, TABLE 2-4, Design of Machinery, 6 Edition

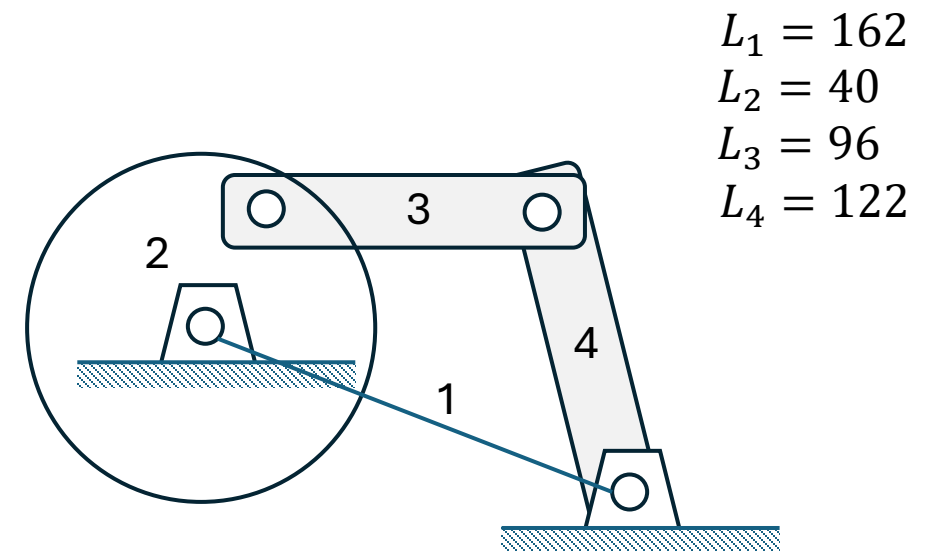
Type	$S + L$ vs $P + Q$	Inversion	Class	Barker's Designation	Code	Also known as
1	<	$L_1 = S = \text{ground}$	I-1	Grashof crank-crank-crank	GCCC	Double-crank
2	<	$L_2 = S = \text{input}$	I-2	Grashof crank-rocker-rocker	GCRR	Crank-rocker
3	<	$L_3 = S = \text{coupler}$	I-3	Grashof rocker-crank-rocker	GRCR	Double-rocker
4	<	$L_4 = S = \text{output}$	I-4	Grashof rocker-rocker-crank	GRRC	Rocker-crank
5	>	$L_1 = L = \text{ground}$	II-1	Class 1 rocker-rocker-rocker	RRR1	Triple-rocker
6	>	$L_2 = L = \text{input}$	II-2	Class 2 rocker-rocker-rocker	RRR2	Triple-rocker
7	>	$L_3 = L = \text{coupler}$	II-3	Class 3 rocker-rocker-rocker	RRR3	Triple-rocker
8	>	$L_4 = L = \text{output}$	II-4	Class 4 rocker-rocker-rocker	RRR4	Triple-rocker
9	=	$L_1 = S = \text{ground}$	III-1	Change-point crank-crank-crank	SCCC	SC* double-crank
10	=	$L_2 = S = \text{input}$	III-2	Change-point crank-rocker-rocker	SCRR	SC crank-rocker
11	=	$L_3 = S = \text{coupler}$	III-3	Change-point rocker-crank-rocker	SRCR	SC double-rocker
12	=	$L_4 = S = \text{output}$	III-4	Change-point rocker-rocker-crank	SRRC	SC rocker-crank
13	=	Two equal pairs	III-5	Double change point	S2X	Parallelogram or deltoid
14	=	$L_1 = L_2 = L_3 = L_4$	III-6	Triple change point	S3X	Square

Note: * SC represent "Special Case".

Barker's Classification Example

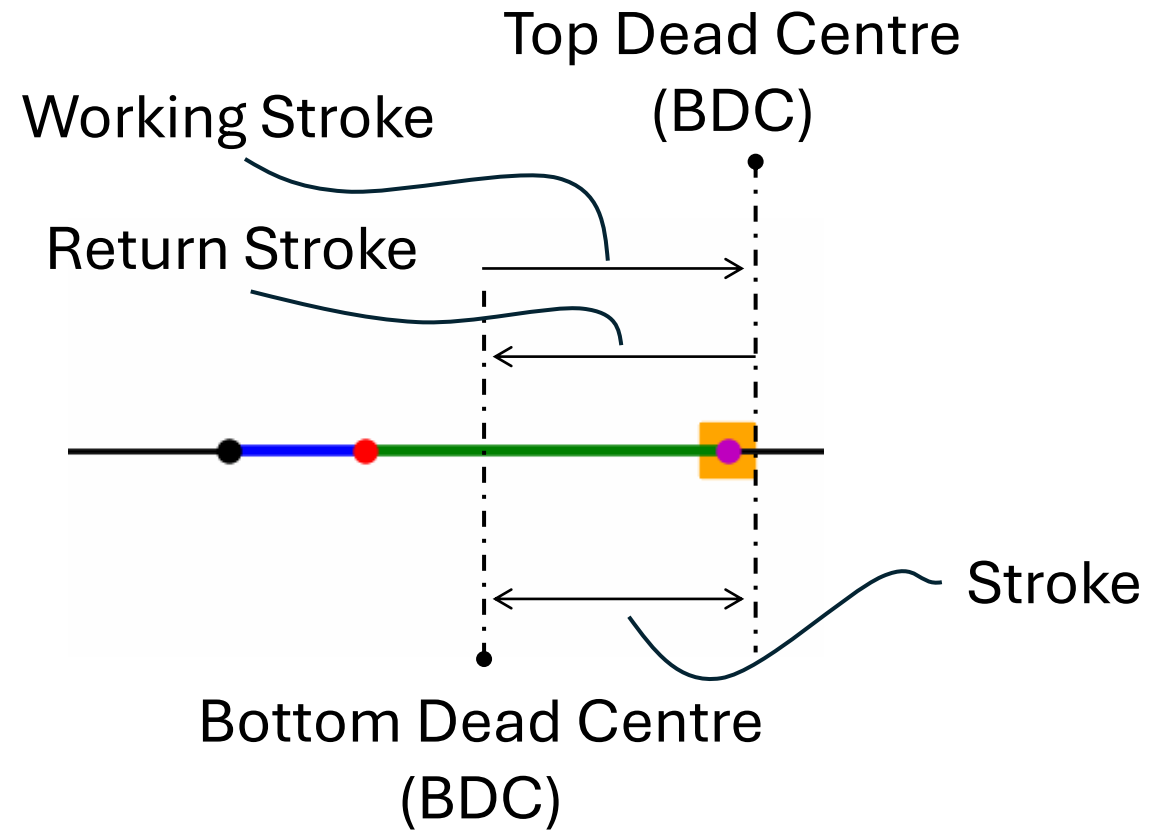
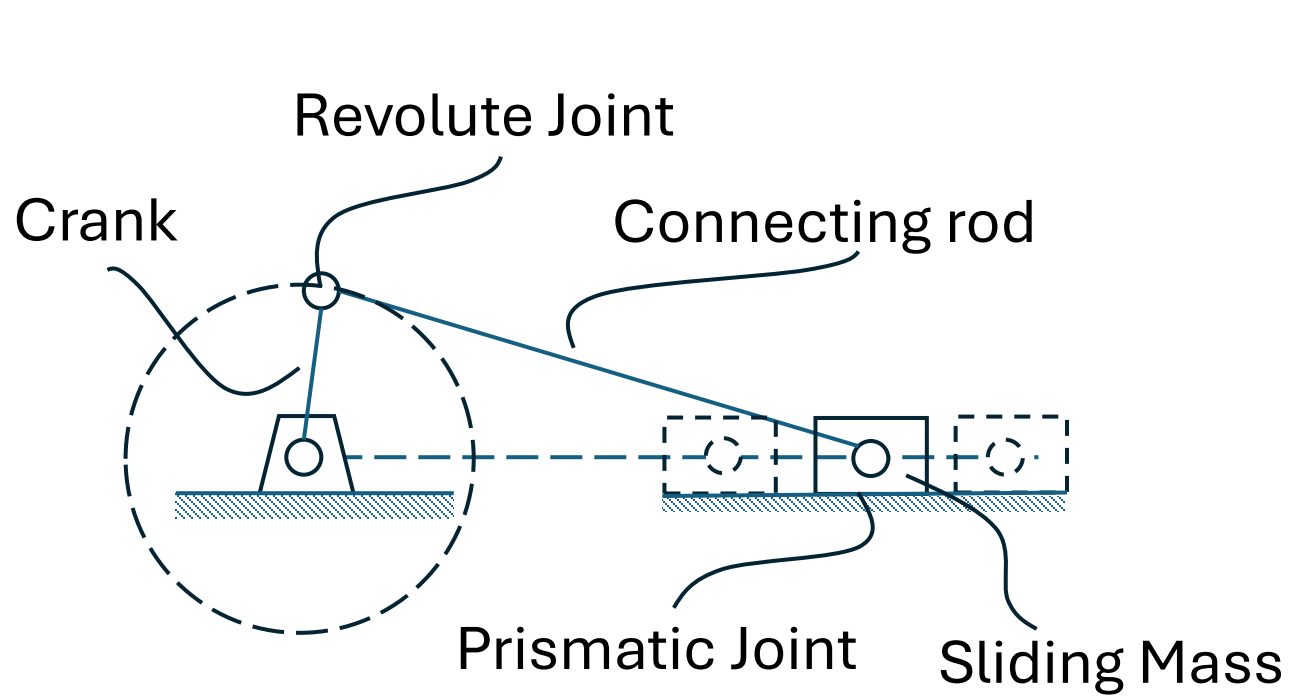


Grashof Condition: _____
 Barker's Classification: _____



Grashof Condition: _____
 Barker's Classification: _____

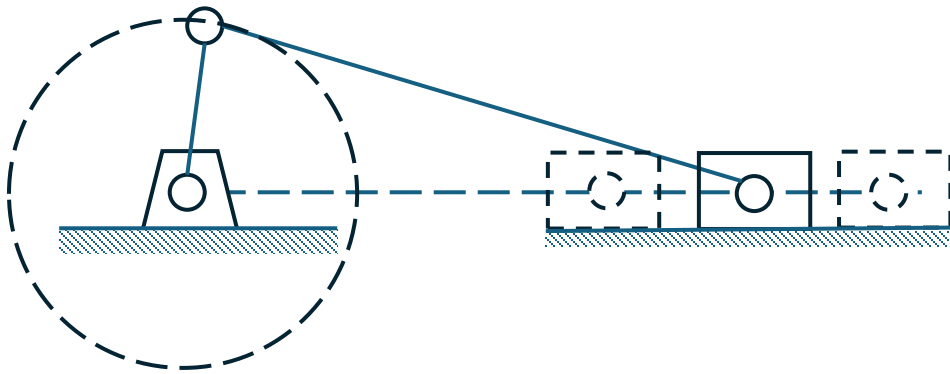
Slider-Crank Mechanism



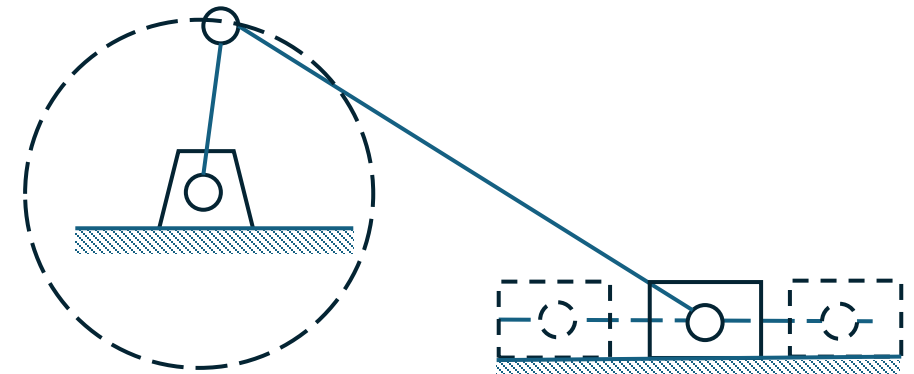
Slider-Crank Mechanism (Cont.)

- A slider crank is most widely used to convert reciprocating to rotary motion (as in an engine) or to convert rotary to reciprocating motion (as in pumps or compressors).
- Positions at which slider motion reverses are called _____.
- When crank and connecting rod are extended in a straight line and the slider is at its maximum distance from the axis of the crankshaft, the position is ____ Dead Centre.
- When the slider is at its minimum distance from the axis of the crankshaft, the position is ____ Dead Centre.

Slider-Crank Mechanism (Cont.)



In-line
Slider-Crank Mechanism



Off-set
Slider-Crank Mechanism